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Case 3-2024: An 8-Week-Old Male Infant with Inconsolable Crying and Weakness

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PRESENTATION OF CASE

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Dr. Jeffrey M. Sumner (Pediatrics): An 8-week-old male infant was admitted to the pediatric intensive care unit (PICU) of this hospital because of inconsolable crying, lethargy, weakness, and hypoxemia.

The patient had been well until 7 days before the current presentation, when irritability and frequent crying developed. One day before the current presentation, the episodes of crying increased in duration, and the patient's grandmother noticed that he cried more intensely when the right side of his abdomen was touched. The patient was evaluated at the pediatric primary care clinic of another hospital. The vital signs and physical examination were reportedly normal, and a diagnosis of discomfort due to gastrointestinal gas was considered.

After the patient returned home, he had a crying episode that lasted for multiple hours while he was awake; the crying stopped while he was fully asleep. After he woke up, the crying continued for 8 hours and became weaker. During breast-feeding, the patient fed less vigorously than usual, and the latch onto the nipple was weaker. That night, the crying continued, and the patient became inconsolable. He had frantic movements of the arms and legs and slept only 1 hour. On the day of the current presentation, the inconsolable crying continued. The patient did not latch onto the nipple, and his mother expressed breast milk directly into his mouth. A family member called the pediatric primary care clinic and was instructed to bring the patient to the emergency department of this hospital for evaluation.

In the emergency department, additional history was obtained from the patient's family members. There had been no change in the frequency of wet diapers; the last bowel movement had occurred 2 days before this presentation. The patient had been born at a gestational age of 39 weeks 1 day by a normal spontaneous vaginal delivery after a normal pregnancy. The birth weight had been 4.5 kg (98th percentile), the length 54.6 cm (99th percentile), and the head circumference 36.5 cm (94th percentile). The patient had met normal developmental milestones. He had not yet received the routine immunizations recommended for 2-month-old infants. Medications included vitamin D supplements; there were no known medica-

tion allergies. The patient lived in an urban area of New England with his grandmother, who primarily cared for him, as well as his mother, father, and sister. His mother had depression, anxiety, and anemia. His father and sister were healthy.

On examination, the axillary temperature was 37.2°C, the blood pressure 100/76 mm Hg, the heart rate 153 beats per minute, the respiratory rate 34 breaths per minute, and the oxygen saturation 97% while the patient was breathing ambient air. The weight was 7.1 kg (99th percentile), the height 62.5 cm (>99th percentile), and the head circumference 41.5 cm (99th percentile). The patient was alert but irritable. He cried weakly and could not be soothed. The anterior fontanelle was open, soft, and flat. The mucous membranes were moist. The heart rate was tachycardic and regular. The lungs were clear on auscultation. The abdomen was soft, with no hepatosplenomegaly. The results of the testicular examination were normal. There was no rash.

The blood levels of glucose, lactic acid, lipase, aspartate aminotransferase, alanine aminotransferase, and bilirubin were normal, as were the results of kidney-function tests. The platelet count was 621,000 per microliter (reference range, 150,000 to 400,000); the remainder of the complete blood count and differential count was normal. The results of urinalysis and urine toxicology screening were normal. Tests for severe acute respiratory syndrome coronavirus 2, influenza A and B viruses, and respiratory syncytial virus were negative; other laboratory test results are shown in Table 1. Blood was obtained for culture. Radiography of the chest and abdomen and ultrasonography of the abdomen showed no abnormalities.

While the patient was in the emergency department, he remained irritable and did not attain a calm awake state. He was admitted to this hospital. Four hours after presentation, lethargy developed. Additional testing was performed. Magnetic resonance imaging of the head showed no abnormalities. A lumbar puncture was performed for cerebrospinal fluid (CSF) analysis. The CSF glucose level was 54 mg per deciliter (3.0 mmol per liter; reference range, 50 to 75 mg per deciliter [2.8 to 4.2 mmol per liter]), and the CSF protein level was 104 mg per deciliter (reference range, 5 to 55); there were 1055 red cells per microliter and 2 nucleated cells per microliter, of which 70% were lymphocytes. A diagnosis of men-

Table 1. Laboratory Data.*

Variable	Reference Range†	On Admission
White-cell count (per μ l)	5000–19,500	5480
Hemoglobin (g/dl)	10.0–18.0	12.8
Hematocrit (%)	31.0–55.0	37.5
Platelet count (per μ l)	150,000–450,000	621,000
Sodium (mmol/liter)	135–145	136
Potassium (mmol/liter)	3.4–5.0	5.4
Chloride (mmol/liter)	98–106	102
Carbon dioxide (mmol/liter)	22–27	18
Urea nitrogen (mg/dl)	5–20	8
Creatinine (mg/dl)	0.30–1.00	0.27
Glucose (mg/dl)	70–110	108
Calcium (mg/dl)	8.5–10.5	10.8
Lactic acid (mmol/liter)	0.5–2.0	1.8
Aspartate aminotransferase (U/liter)	9–80	43
Alanine aminotransferase (U/liter)	10–55	26
Alkaline phosphatase (U/liter)	122–469	658
Total bilirubin (mg/dl)	<1.2	3.1
Albumin (g/dl)	3.3–5.0	4.8
Globulin (g/dl)	1.9–4.1	1.8
Total protein (g/dl)	6.0–8.3	6.6

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† The ranges used at Massachusetts General Hospital are for children who do not have medical conditions that could affect the results. They may therefore not be appropriate for all neonates.

ingitis was considered. Empirical treatment with intravenous vancomycin, ceftriaxone, and acyclovir was started.

Six hours after presentation, increased work of breathing developed, and the oxygen saturation decreased to 85% while the patient was breathing ambient air. Supplemental oxygen was administered through a nasal cannula at a rate of 2 liters per minute, and the oxygen saturation increased to 100%. The patient was transferred to the PICU.

Dr. Sarah A. Murphy: On arrival at the PICU, the patient was difficult to arouse. He had grunting respirations, head-bobbing, intercostal retractions, and an exaggerated abdominal breathing pattern. With vigorous stimulation, the patient had a weak hypophonic cry, gurgling, and stertor on inspiration. Ptosis was present in both eyes. When

the eyes were opened by the examiner, the pupils were equal, round, and reactive. The face appeared symmetric and placid. There were few spontaneous movements of the arms, and the patient did not move his arms against gravity. There was no spontaneous movement of the legs, and diffuse hypotonia was present. Muscle bulk was normal and symmetric. Reflexes were decreased but preserved.

Tracheal intubation was performed, and mechanical ventilation was initiated. Tachycardia and hypertension persisted despite the administration of boluses of sedative and analgesic agents and the initiation of continuous infusions of opioids and benzodiazepines.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Adam Berkowitz: This previously healthy 8-week-old male infant presented with irritability that had lasted for 1 week, a frequently encountered clinical presentation. After he was admitted to the hospital, lethargy and hypotonia developed rapidly. It is likely that the patient's irritability, lethargy, and hypotonia have a unifying cause. I will begin by sharing my approach to the evaluation of an infant with irritability. I will then attempt to narrow the differential diagnosis by considering causes of hypotonia.

IRRITABILITY IN AN INFANT

Irritability in an infant can be caused by common, easily reversible conditions but also by life-threatening conditions that require rapid recognition and treatment. Symptoms may not be clear in infants, and signs can be easily missed. A systematic approach that involves an evaluation of each body system sequentially can be helpful.¹ However, given the immunocompromised and unimmunized status of infants, I will first consider infections.

Infections

A localized infection in an infant can easily spread to cause an invasive disease, such as bacteremia or meningitis. This patient's examination findings ruled out common infections, such as acute otitis media and skin and soft-tissue infections, that may have served as initial foci of invasive disease. The normal results on urinalysis lower the likelihood of a urinary tract infection. The presence

of only 2 nucleated cells per microliter on CSF analysis is not consistent with meningitis. The findings on chest radiography are not suggestive of pneumonia. He did not have pseudoparalysis of an arm or leg, which would suggest underlying osteomyelitis or an infected joint. In addition, he did not have fever or hypothermia, and the white-cell count was normal. Because there was no clear evidence of an infection that would explain the patient's irritability, I will next use the head-to-toe method to consider causes of irritability associated with each system of the body.

Neurologic Causes

Neurologic causes of irritability in an infant — including stroke, hydrocephalus, cerebral edema, and masses — are unlikely in this patient, given that he did not have focal neurologic deficits or a bulging fontanelle and had normal results on neuroimaging. Seizures are also unlikely because he did not have any witnessed seizure activity.

Ocular Causes

Pain from a corneal abrasion can cause irritability in an infant. A fluorescein test could be considered to further evaluate for this possibility.

Cardiopulmonary Causes

The patient's normal chest radiograph, without evidence of cardiomegaly or pulmonary edema, makes heart failure and myocarditis unlikely. An arrhythmia is also unlikely, given the normal heart rate. There is no report of foreign-body ingestion, choking, or abnormal respiratory sounds that would suggest aspiration.

Gastrointestinal Causes

Parents and pediatricians frequently anchor on the gastrointestinal system when trying to find a reason for irritability. Common diagnoses include constipation, gas, reflux, and colic. This patient's irritability was initially attributed to gas when he was first evaluated by the primary care physician. When he presented to this hospital, he had no history of bilious emesis, and the findings on abdominal examination, the abdominal radiograph and ultrasound image, and the blood levels of lipase, aspartate aminotransferase, alanine aminotransferase, and bilirubin were all normal. These features make appendicitis, intussusception, pancreatitis, hepatobiliary disease, and other causes of obstruction unlikely. Furthermore,

appendicitis and intussusception occur rarely in this age group.

Genitourinary Causes

The patient's normal findings on testicular examination make testicular torsion, hernia, or hair tourniquet unlikely to be the cause of his irritability. The normal results on urinalysis are not consistent with nephrolithiasis.

Skin and Soft-Tissue Causes

To rule out irritability associated with pain, an examination of all parts of the skin and musculoskeletal system for any signs of injury is essential. Fractures, which can be accidental or possibly signify child abuse, may be associated with swelling, bruising, or decreased movement, none of which was present when the patient was initially evaluated at this hospital.

Cancer

Infants with leukemia or neuroblastoma can present with irritability. The absence of fever, hepatomegaly, cytopenia, and peripheral blasts in this patient is reassuring. Furthermore, the absence of opsoclonus–myoclonus syndrome and ecchymoses around the eyes makes neuroblastoma unlikely, as do the normal abdominal ultrasound image and chest radiograph.

Metabolic Causes

Acute dehydration may be associated with irritability and was probably present in this patient, given the reported decreased oral intake and the decreased bicarbonate level with an elevated anion gap, which was presumably from ketosis, since he had a normal lactic acid level. Congenital conditions involving inborn errors of metabolism are important considerations in an infant with irritability. This patient had been healthy and developing appropriately, so these diagnoses are unlikely. Furthermore, he did not have evidence of hepatomegaly or dysmorphisms on examination, and the absence of marked abnormalities in electrolyte levels and acid–base status also reassures that there is not an underlying congenital metabolic cause.

Ingestions and Toxidromes

Pediatricians would consider ingestions and toxidromes in any patient with irritability. However, these conditions are more likely to occur in tod-

dlers. An ingestion in this 8-week-old infant is unlikely, and urine toxicology screening was negative.

HYPOTONIA IN AN INFANT

Despite the use of a systematic approach, there is no obvious cause of irritability on the basis of the initial patient history and the findings on physical examination, laboratory testing, and imaging. I will now focus on the differential diagnosis of hypotonia in this patient. The evaluation of an infant with hypotonia centers on conditions that affect either the central nervous system (CNS), including the brain and spinal cord, or the peripheral nervous system (PNS), including the motor neurons, muscles, nerves, and neuromuscular junction.² Two rare causes of hypotonia that do not quite fit into this physiologic model are tick paralysis and paralytic shellfish poisoning, neither of which is likely in this patient, given that no ticks were observed on examination and no history of shellfish ingestion was reported.

CNS Disease

In infants, 60 to 80% of cases of hypotonia are attributed to CNS diseases, specifically hypoxic–ischemic encephalopathy and cerebral palsy.³ However, this patient's normal results on neuroimaging and reduced reflexes make both of these diagnoses unlikely. Imaging of the spinal cord was not performed, but there is no report of trauma that would suggest spinal cord injury, and the hypotonia was generalized. Other infectious or postinfectious conditions — such as transverse myelitis, acute disseminated encephalomyelitis, or acute flaccid myelitis — are unlikely in the absence of fever, a preceding illness, and CSF pleocytosis.

PNS Disease

In this patient with normal results on neuroimaging and with features suggestive of bulbar palsies (a weak cry, ptosis in both eyes, and poor feeding), a PNS disease is the most likely cause of hypotonia. The overall severe and acute presentation, which occurred after previous normal development, makes congenital or metabolic myopathy unlikely, as does the normal muscle bulk on examination. An acquired myositis would be unlikely to cause such a severe presentation and would not explain the patient's altered mental status. Spinal muscular atrophy, which is a degen-

erative motor neuron disease, has a rapid onset of symptoms in rare cases. However, no other musculoskeletal abnormalities or tongue fasciculations that would suggest this diagnosis were described in this case.

Could a disease affecting the nerves explain this patient's presentation? Congenital demyelinating conditions are unlikely, given the tempo of illness in this previously healthy infant. The Guillain-Barré syndrome is an acquired demyelinating syndrome that is typically associated with ascending weakness, a loss of reflexes, and an elevated CSF protein level without pleocytosis. Although this patient's CSF protein level was elevated, the Guillain-Barré syndrome is extremely rare in this age group.

Diseases of the neuromuscular junction can also cause hypotonia. There was no reported maternal history of myasthenia gravis that would support the possibility of congenital myasthenia gravis. Transient forms of myasthenia gravis are extremely rare in infants. The Lambert-Eaton myasthenic syndrome, a paraneoplastic process, is also quite rare in infants.

Botulism

Infant botulism is a disease of the neuromuscular junction that fits very well with this patient's presentation. He had many of the associated signs and symptoms, including irritability, constipation, features of bulbar palsies (a weak cry, ptosis in both eyes, and poor feeding), lethargy, weakness, and respiratory difficulties.⁴ Patients with infant botulism typically present with manifestations at 2 to 26 weeks of age. Although infant botulism is known to be associated with honey ingestion, most patients do not have a reported honey exposure; the disease is more likely to result from environmental factors.⁵

There is one aspect of this patient's presentation that would not be explained by a diagnosis of infant botulism: the elevated CSF protein level. An elevated CSF protein level is a feature of the Guillain-Barré syndrome, and the patient had other findings consistent with this condition, including ascending weakness and a loss of reflexes. However, the Guillain-Barré syndrome tends to be diagnosed in older children, whereas approximately 95% of confirmed cases of botulism occur in patients younger than 6 months of age.⁵ An alternative explanation for the elevated

CSF protein level is a traumatic lumbar puncture, which occurs with 10 to 30% of attempted lumbar punctures and could also explain the elevated red-cell count in the CSF in this patient.⁶

Overall, I favor a diagnosis of infant botulism. I would send a stool specimen to be tested for *Clostridium botulinum* toxin type A and would initiate empirical treatment with human-derived anti-botulism toxin antibodies (infant botulism immune globulin, known as BabyBIG) while awaiting the test results.

DR. ADAM BERKWITT'S DIAGNOSIS

Infant botulism.

DIAGNOSTIC TESTING

Dr. Murphy: After the patient was transferred to the PICU, a diagnosis of infant botulism was suspected, and a stool specimen was obtained for botulinum neurotoxin testing. The Infant Botulism Treatment and Prevention Program of the California Department of Public Health was contacted for the BabyBIG antibody treatment. Further clinical history was also obtained.

On further interviewing, the patient's family members reported that he typically had hard stools every 2 to 3 days. Two days before admission, the infant appeared to have abdominal discomfort, which his family members presumed was from constipation or gas. Honey was given to try to soothe him.

Real-time polymerase-chain-reaction testing performed at the Wadsworth Center of the New York State Department of Health was positive for the gene encoding *C. botulinum* toxin type A. This test result confirmed the diagnosis of infant botulism.

MOLECULAR DIAGNOSIS

Clostridium botulinum toxin type A.

DISCUSSION OF INFECTIOUS DISEASE MANAGEMENT

Dr. Chadi M. El Saleeby: The neurotoxins produced mostly by *C. botulinum* are the drivers of the clinical illness known as botulism. There are multiple *C. botulinum* transmission categories: the bacte-

rial spores or toxins can be introduced into the body through food consumption (infant botulism, foodborne botulism, or adult intestinal toxemia), through contamination of an infected wound (wound botulism), or iatrogenically (botulism associated with cosmetics or migraine treatment). Botulinum neurotoxins have also been identified as potential agents for bioterrorism. Overall, botulism is a rare disease in the United States, and infant botulism is the most common transmission category.⁷

In patients with infant botulism, the ingestion of spores may result in colonization and germination, followed by toxin production and subsequent toxemia. Clearly defined food exposures, such as exposures to honey or corn syrup, account for only a minority of cases.⁸ In fact, multiple factors affect the acquisition of the clinical disease, with infants' guts being particularly susceptible. Often, there is a history involving rural living, dust production, or nearby soil perturbation.⁹ The incubation period after exposure to spores may be a few days (as seen in this patient) to a few weeks.

After sporulation, the enteric botulinum neurotoxin enters the systemic circulation and interrupts the normal signal transmission at the neuromuscular junction (Fig. 1). The enteric botulinum neurotoxin irreversibly binds a specific receptor on the presynaptic peripheral cholinergic nerve terminal and then enters the cell, breaking down a critical protein associated with exocytosis and interrupting the release of the neurotransmitter acetylcholine into the intersynaptic space. Once cleaved, this exocytosis-associated protein complex requires up to 4 weeks to regenerate, which explains the protracted course of the illness. BabyBIG antibodies do not penetrate the neuron, so it is critical to administer the antitoxin as soon as possible, before the enteric botulinum neurotoxin enters the cell.

The diagnosis of botulism is typically made with the detection of the enteric botulinum neurotoxin in stool. This test is generally available only through local departments of public health, and it may take several days for the results to become available. Therefore, it is not appropriate to delay the administration of the antitoxin while waiting for the diagnosis to be microbiologically confirmed. Early recognition of the illness and prompt administration of medication, preferably

within 7 days after symptom onset, are imperative to increase the likelihood of a good outcome.

DISCUSSION OF MANAGEMENT IN THE PICU

Dr. Murphy: In the treatment of this infant with suspected botulism, the priority was to assess and support the airway, breathing, and circulation. The neuromuscular paralysis associated with botulism can lead to mechanical respiratory failure by affecting the muscles involved in respiration (respiratory pump failure), the muscles involved in maintaining the airway (loss of airway tone), or the muscles involved in airway clearance (poor cough). Tracheal intubation and mechanical ventilation are indicated in 50 to 80% of infants who are hospitalized with botulism,¹⁰⁻¹³ and the majority of these patients have progression to invasive ventilatory support within 24 hours after presentation.¹⁰

Respiratory failure due to neuromuscular weakness follows a typical progression,¹⁴ with decreased tidal volumes and decreased vital capacity causing progressive atelectasis and stimulating tachypnea. Atelectasis results in further alveolar hypoventilation and ventilation-perfusion mismatch and in a further increase in respiratory work. Poor airway clearance and loss of airway tone add to the workload that must be overcome by the respiratory muscles. The resulting increased workload puts additional strain on fatiguing respiratory muscles, leading to worsening of atelectasis, and respiratory decompensation worsens (Fig. 2).

Data regarding clinical predictors of respiratory failure in patients with botulism are limited, and hypoxemia and hypercarbia are late manifestations of respiratory failure. Therefore, the decision to perform tracheal intubation and initiate mechanical ventilation should be based on clinical judgment. An assessment of respiratory decompensation must also be considered in the context of attendant neuromuscular weakness. Bulbar and facial muscle weakness may mask the clinical signs typically associated with respiratory distress, such as facial grimacing or nasal flaring. In the context of diaphragm paralysis, a paradoxical abdominal breathing pattern may be observed.

At the time of the patient's arrival at the PICU,

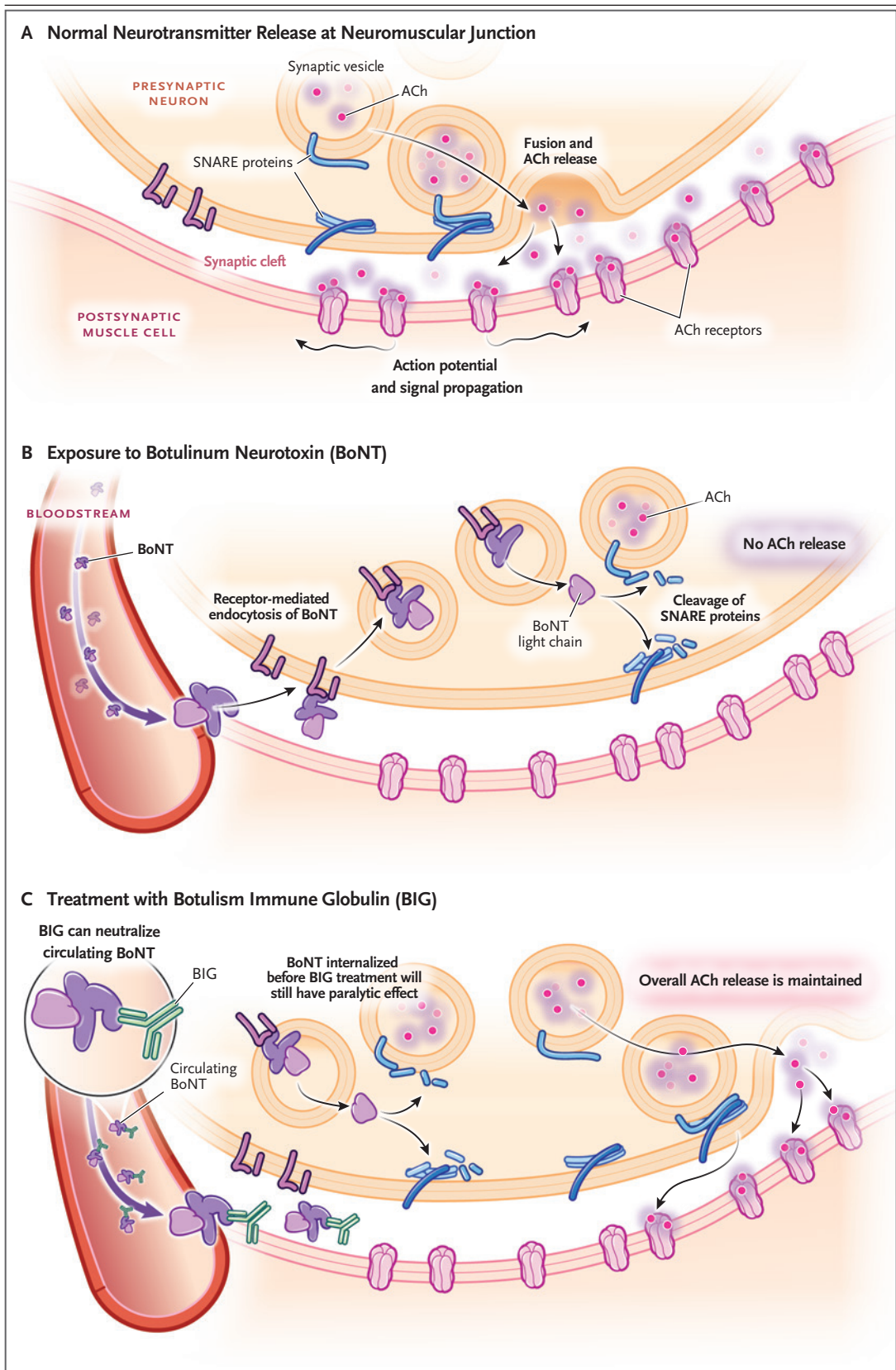


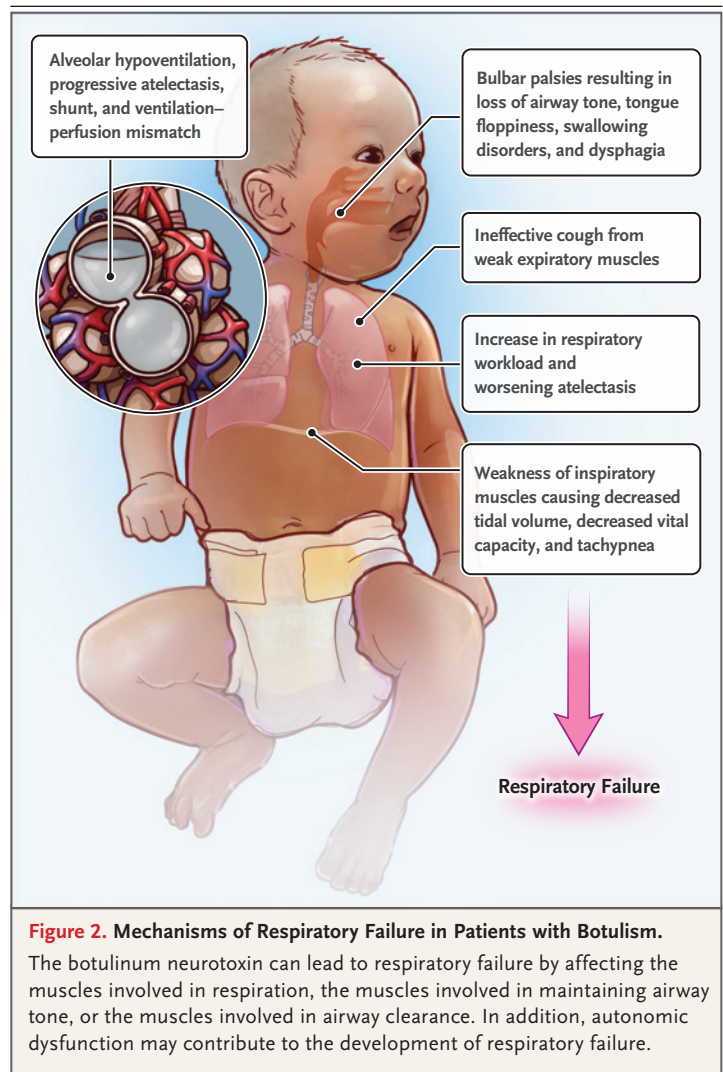
Figure 1 (facing page). Action of the Botulinum Neurotoxin.

During normal function (Panel A), the vesicle of the presynaptic nerve cell that contains acetylcholine (ACh) binds the nerve cell membrane, a process mediated by the soluble *N*-ethylmaleimide-sensitive factor attachment protein receptor (SNARE) protein complex. After exocytosis, ACh is released into the synaptic cleft of the neuromuscular junction and binds the ACh receptor of the postsynaptic muscle cell, causing a change in action potential and signal propagation through the muscle cell. In a patient with botulism (Panel B), the botulinum neurotoxin (BoNT) enters the presynaptic nerve cell through endocytosis, by binding to the BoNT receptor. The BoNT light chain then passes into the cytosol of the nerve cell, cleaving the SNARE proteins and preventing ACh vesicle fusion and neurotransmitter release. Treatment with botulism immune globulin, known as BIG (Panel C), neutralizes circulating BoNT. BIG is not effective once the toxin has entered the presynaptic nerve cell; therefore, BIG cannot reverse existing weakness and paralysis.

he had features indicative of respiratory failure due to progressive neuromuscular weakness. Findings consistent with respiratory distress included increased work of breathing, an increased respiratory rate, the use of accessory muscles of respiration, and signs of an acute stress response, such as agitation and irritability, tachycardia, and hypertension. In addition to oxygen desaturation and the inability to protect the airway, with stertorous breath sounds and secretions, the patient had altered mental status, which in this context may be due to acute respiratory failure.

The botulinum neurotoxin binds cholinergic receptors, affecting both motor and autonomic function. Autonomic disturbances are common, and tachycardia, hypertension, and flushing are frequently reported in hospitalized infants with botulism.^{11,15} This patient had tachycardia and hypertension after tracheal intubation that persisted despite the administration of boluses of sedative and analgesic agents. Hypertension and tachycardia can be treated with short-acting agents, but such treatment was not considered to be necessary in this case. Patients are monitored for autonomic instability, given that the resolution of autonomic disturbances may be delayed.¹⁶

The BabyBIG antibody treatment arrived from California and was administered to the patient 31 hours after presentation, on the basis of a strong clinical suspicion of infant botulism and before the diagnosis was confirmed. Six days af-

**Figure 2. Mechanisms of Respiratory Failure in Patients with Botulism.**

The botulinum neurotoxin can lead to respiratory failure by affecting the muscles involved in respiration, the muscles involved in maintaining airway tone, or the muscles involved in airway clearance. In addition, autonomic dysfunction may contribute to the development of respiratory failure.

ter presentation, test results showing the detection of *C. botulinum* toxin type A in the stool were reported, confirming the diagnosis of infant botulism.

Tracheal intubation was maintained until hospital day 10, when the patient regained movement of his arms and legs. He was able to lift his arms off the bed, spontaneously open his eyes, and cough. Before extubation, the patient was able to generate appropriate tidal volumes and did not have signs of respiratory distress during a trial performed with the use of continuous positive airway support. After extubation, he received non-invasive ventilatory support for a brief transition period. The patient continued to be fed through a nasogastric tube after successful extubation. He had ongoing hypotonia and difficulty with feed-

ing and was discharged to a rehabilitation facility on hospital day 21. He spent 8 days in the rehabilitation facility before being discharged home.

When the patient was seen in the outpatient pediatric neurology clinic 3 months later, he had fully recovered. He had a normal physical examination and was developmentally on track. At his recent 3-year check-up, he continued to do well.

FINAL DIAGNOSIS

Infant botulism.

This case was presented at the Yale New Haven Health postgraduate course “New England Pediatric Hospital Medicine 2023,” directed by Drs. Magna Dias, Chadi El Saleeby, Jaspreet Loyal, Kathryn Niro, Daniel Rauch, and Kerstin Zanger.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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